Paper OOL-9: Protocell Integration, Replication, and Error Thresholds: The First Heritable Container Gate

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May 2026

Abstract

This paper is the integration gate. A protocell claim requires bounded copying error, retained internal chemistry, coupled energy throughput, and a boundary whose persistence covaries with its internal state. The paper combines Eigen–Orgel error-threshold logic with a conservative protocell diagnostic and explicitly reports numerical fragility where it appears. That restraint is intentional: Paper 9 defines what must be true before Paper 10 can introduce parasitic exploitation.

Symbol Conventions

CDFD origins-of-life symbol convention.

Symbol	Meaning in Part II	Origins-of-life reading
Φ	Available driving flux	Redox, thermal, photochemical, proton/ion, or chemical potential drive supplied by the environment
C	Effective constraint or resistance	Mineral geometry, pore confinement, diffusion barrier, permeability, activation barrier, or maintenance burden
S	Responsive organization	Surface, boundary, catalytic network, coacervate, or protocellular structure that routes flux into retained function
M_s	Retained structural memory	Composition, sequence, topology, compartment state, mineral imprint, or boundary history that persists across renewal cycles
Ψ_s	Local adaptive ratio $(\Phi/C) \cdot S \cdot M_s$	Bookkeeping gauge for whether drive, constraint, responsiveness, and retention are co-present; not a universal constant
Λ_{life}	Integrated Life Number where used	Capstone surplus ratio for decay, near-critical proto-biology, and sustained self-maintenance

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Accumulation or reinforcement coefficient	Defines how fast a constraint, pathway, or retained structure grows under load
β	Relaxation, decay, or washout coefficient	Defines recovery, loss, clearance, hydrolysis, or return toward baseline
γ	Coupling, diffusion, or redistribution coefficient	Defines spread across pores, surfaces, compartments, networks, or chemical phases
δ, ϵ	Perturbation, tolerance, or small offset scales	Used only as local thresholds, numerical guards, or experimental tolerances
η, λ, μ, ν	Efficiency, rate, baseline, or frequency terms	Meaning is set by the equation and subscript where they occur
ρ, σ, τ, ϕ	Density/correlation, conductance/noise, time-scale, or phase/-field terms	Never universal constants unless a paper explicitly defines them that way
Ω, Λ	Domain/state space; Life Number where used	Ω names a modeled state space; Λ is reserved for life-number notation only where defined

1 Introduction

Papers 1–8 build the prerequisite ladder: thermodynamic framing, geochemical energy and transport, oscillating polymerization, autocatalytic closure, and adaptive boundaries. A protocell requires these pieces to become coupled. Replication without a boundary may preserve molecular information but does not necessarily create a lineage. A boundary without inherited internal chemistry can divide physically without biological heredity.

Paper 9 therefore asks two linked questions:

1. When does copying preserve information rather than dissolve into error?
2. When does copying affect the persistence and division of the container that carries it?

2 Template Growth and Replication Gain

For a schematic template T and monomer pool M_i,

$$T + M_i \to TM_i, \qquad R_{\text{rep}} = k_{\text{rep}} c_T c_M. \tag{1}$$

Transport-limited monomer supply can be represented phenomenologically as

$$\boldsymbol{J}_M = -\frac{1}{C} \nabla c_M. \tag{2}$$

A well-mixed caricature gives

$$\frac{dc_T}{dt} = k_{\text{rep}} c_T c_M - \frac{1}{CL^2} c_T = c_T \left(k_{\text{rep}} c_M - \frac{1}{CL^2} \right). \tag{3}$$

Define the replication gain

$$\Gamma_{\text{rep}} = k_{\text{rep}} c_M C L^2. \tag{4}$$

The condition $\Gamma_{\text{rep}} > 1$ is only a linearized growth condition. It is not sufficient for life.

3 Error Threshold

If per-step fidelity is q across n additions, the simplest master-sequence survival approximation is

$$Q = q^n, \qquad \epsilon = 1 - Q. \tag{5}$$

Information persists only if replication gain outruns error catastrophe. A schematic compatibility condition is

$$\Gamma_{\text{rep}} Q > 1, \tag{6}$$

with the usual warning that real prebiotic copying is not a clean independent-site process [4, 2].

The CDFD hypothesis is that fidelity, monomer activation, retention, and damage are coupled to energy throughput. A useful bookkeeping score is

$$\mathcal{H}_{\text{copy}} = \Gamma_{\text{rep}} Q R_{\text{mono}} R_{\text{ret}} R_{\text{damage}}^{-1}, \tag{7}$$

where R_{mono} is activated-monomer availability, R_{ret} is retention of templates and products, and R_{damage} collects hydrolysis, photodamage, side reactions, and strand loss. This is a checklist, not a calibrated law.

4 Quasispecies Logic Versus Protocell Logic

The quasispecies threshold and the protocell threshold are related but not identical. The first asks whether a sequence distribution preserves information. The second asks whether a bounded domain inherits a coupled chemistry that affects its growth or persistence.

The protocell condition can be stated as a covariance requirement:

$$\mathrm{Cov}(\text{copied internal state}, \text{container persistence}) > 0. \tag{8}$$

Without this covariance, selection may act on free templates or on empty containers, but not yet on a protocell lineage.

5 Integrated Protocell Scaffold

An integrated toy protocell has state variables such as

$$U = \{c_i, \Phi_e, \Phi_H, C_i, \ldots\}, \tag{9}$$

evolving through transport, reaction, and constraint laws [6]. A minimal transport skeleton is

$$\frac{\partial c_i}{\partial t} = \nabla \cdot \left(\frac{1}{C_i} \nabla c_i \right) + R_i(c, \Phi_e, \Phi_H), \tag{10}$$

with potentials satisfying analogous updates:

$$\frac{\partial \Phi_e}{\partial t} = \nabla \cdot \left(\frac{1}{C_e} \nabla \Phi_e \right) + S_e, \tag{11}$$

$$\frac{\partial \Phi_H}{\partial t} = \nabla \cdot \left(\frac{1}{C_H} \nabla \Phi_H \right) + S_H. \tag{12}$$

Flux-coupled reactions can be written as

$$\boldsymbol{J}_e = -\frac{1}{C_e} \nabla \Phi_e, \qquad \boldsymbol{J}_H = -\frac{1}{C_H} \nabla \Phi_H, \tag{13}$$

$$R_i = f_i(c)\, g(|\boldsymbol{J}_e \cdot \boldsymbol{J}_H|), \tag{14}$$

for nonnegative g.

6 Numerical Audit

The diagnostic for this paper combines a reduced error-threshold sweep with a bounded two-dimensional protocell update. It should be read as a numerical stress test, not as validation. The reduced diagnostic reported:

- $\Phi = 0.2$: final master-sequence fitness 0.367,
- $\Phi = 0.5$: final master-sequence fitness 0.083,

- $\Phi = 1.0$: final master-sequence fitness 0.014,
- $\Phi = 5.0$: final master-sequence fitness 1.000,
- $\Phi = 10.0$: final master-sequence fitness 1.000.

The non-monotonic low-to-intermediate response rules out a single universal critical flux in this reduced diagnostic. The correct claim is windowed: copying benefits from energy throughput only when activation, retention, and damage remain bounded.

The final release replaces the earlier unbounded explicit autocatalytic update with saturated kinetics and bounded adaptive S, M_s updates. Under those release parameters, the two-dimensional protocell diagnostic remains finite for the full run. The result should still be read conservatively: the simulated boundary protects a low internal Ψ_s region while the exterior experiences large drive, which is a stress-test picture of compartmental buffering rather than a demonstration of real protocell reproduction.

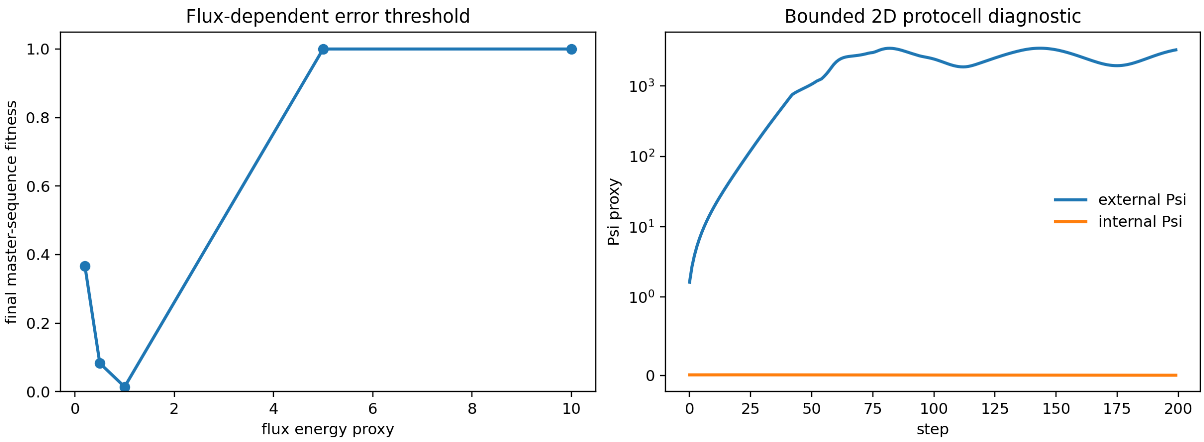

Figure 1: Merged error-threshold/protocell diagnostic. The left panel tests flux-dependent master-sequence persistence. The right panel shows a bounded two-dimensional protocell stress test with finite internal and external adaptive-ratio traces. The figure is a mechanism diagnostic, not a successful protocell simulation.

7 Hardening Roadmap

The release implementation now uses saturated kinetics such as

$$R_{\text{auto}} = \frac{k\Phi^2}{K + \Phi^2} - \ell\Phi, \tag{15}$$

and verifies boundedness in the generated summary JSON. The next hardening step is stronger: timestep refinement, grid-size refinement, and comparison against operator-split or implicit updates for coupled transport and reaction. The model should only be promoted from stress test to prediction if those refinement checks preserve the qualitative result.

8 Experimental Targets

The model protocell literature gives concrete targets: template-directed genetic-polymer synthesis in fatty-acid vesicles [3], nonenzymatic RNA synthesis inside model protocells [1], multi-step nonenzymatic primer-extension chemistry [7], and recent water-compatible thioester-mediated RNA aminoacylation/peptidyl-RNA chemistry [5]. CDFD should therefore be tested by a coupled score,

$$\mathcal{R}_{\mathrm{proto}} = \Gamma_{\mathrm{rep}} Q R_{\mathrm{retention}} R_{\mathrm{exchange}} R_{\mathrm{division}}, \tag{16}$$

not by replication alone.

9 Conclusion

Paper 9 is the integration gate. It establishes that bounded copying, retained internal chemistry, coupled energy flow, and heritable containers must be tested together. The current numerical status is deliberately conservative: the reduced copying toy is informative, while the two-dimensional protocell implementation is a bounded stress test that still needs refinement before it can be treated as a prediction. Paper 10 can only discuss parasitic exploitation after this replication/container machinery exists.

10 Reproducibility Note

The companion script `supplementary_ool_09.py` writes the figure `outputs/paper_09/error_threshold_and_protocell.png` plus CSV/JSON summaries. No notebook is required.

References

[1] Katarzyna Adamala and Jack W. Szostak. Nonenzymatic template-directed RNA synthesis inside model protocells. *Science*, 342:1098–1100, 2013.

[2] Manfred Eigen. Selforganization of matter and the evolution of biological macromolecules. *Naturwissenschaften*, 58:465–523, 1971.

[3] Sheref S. Mansy, Jason P. Schrum, Mathangi Krishnamurthy, et al. Template-directed synthesis of a genetic polymer in a model protocell. *Nature*, 454:122–125, 2008.

[4] Leslie E. Orgel. The maintenance of the accuracy of protein synthesis and its relevance to ageing. *Proceedings of the National Academy of Sciences*, 49:517–521, 1963.

[5] Jyoti Singh, Benjamin Thoma, Daniel Whitaker, et al. Thioester-mediated RNA aminoacylation and peptidyl-RNA synthesis in water. *Nature*, 644:933–944, 2025.

[6] Eric Smith and Harold J. Morowitz. *The Origin and Nature of Life on Earth: The Emergence of the Fourth Geosphere*. Cambridge University Press, 2016.

[7] Travis Walton, Lydia Pazienza, and Jack W. Szostak. Template-directed catalysis of a multistep reaction pathway for nonenzymatic RNA primer extension. *Biochemistry*, 58(6):755–762, 2019.